Practicing Probability

DATA 118 Lab — Student Version

Goals

Recall that in statistics and data science, the idea is to detect patterns in your variables to understand relationships, make predictions, and draw inferences. In order to be able to do so, we need to know how to talk about how variables theoretically behave using probability.

Information can be conveyed about probability in a few different ways. In class, we covered one-way tables, Venn diagrams, and two-way contingency tables–now, we will review those rules while also linking them back to topics we have already covered, including how to calculate probabilities using tools like lists, arrays, and data frames.

The goals of this lab are:

- To practice using the rules of probability, including the General Addition Rule, the Multiplication Rule, and the Complement Rule.
- To practice calculating conditional probabilities.
- To interpret various probability distribution displays, including Venn Diagrams, one-way tables, and contingency tables.
- To reinforce what we know about probability using other tools like data frames.

Setup

For this lab we will be using `math`, `numpy`, and `pandas`.

Exercise

```
import math
import numpy as np
import pandas as pd
```

One-Way Tables

Consider a fictitious university similar to the University of Chicago in many ways–however, a policy of this university is that students may choose one and only one major. We have a data frame with information from 500 of this university's students.

Exercise

```
# To download this file go to
 ↪  https://posit.ds.uchicago.edu/probabilitypracticelab/www/students.csv

student_majors = pd.read_csv("./data/students.csv")
```

Use this cell to carry out some exploratory data analysis using either built-in Python functions or `pandas`-specific methods and attributes. You will need to explore the `student_majors` data frame to answer the following questions.

Exercise — write your code here:

Question 1. How many students are there in the data frame? A.k.a., if we were to calculate probabilities for this data set, what will our denominator be?
Answer:
_____

Question 2. What kind of variable is `major`?
☐ Discrete (Numeric)
☐ Continuous (Numeric)
☐ Nominal (Categorical)

Question 3. Describe the sample space of student majors at this university.
☐ $S = \{0, 1, 2, ..., 497, 498, 499\}$
☐ $S = \{1, 2, 3, ..., 498, 499, 500\}$
☐ $S = \{$ `major`, `DATA118` `DATA119`, `DATA120` $\}$
☐ $S = \{$ `BIOS`, `DATA`, `ECON`, `MATH`, `PLSC` $\}$
☐ $S = \{$ `BIOS`, `DATA`, `ECON`, `MATH`, `PLSC`, `Other` $\}$

Now, use `.groupby()` on `student_majors` to find the counts of each major. Be sure to view on the `"SID"` column, and save your new object as `major_dist`.

Exercise — write your code here:

Now, use this cell, as well as your `major_dist` object, to calculate the probabilities described in the questions below.

Exercise — write your code here:

Question 4. What is the probability a student is a DATA major? Round to three decimal places.
Answer:
_____

Question 5. What is the probability a student is a DATA or an ECON major? Round to three decimal places.
Answer:
_____

Discuss with a neighbor (or on Ed):
How would your approach to answering the previous question change if this university allowed students to have more than one major?

Question 6. What is the complement of the event `major == 'Other'`?
□ $S = \{0, 1, 2, ..., 497, 498, 499\}$
□ $S = \{1, 2, 3, ..., 498, 499, 500\}$
□ $S = \{$ `major`, `DATA118` `DATA119`, `DATA120` $\}$
□ $S = \{$ `BIOS`, `DATA`, `ECON`, `MATH`, `PLSC` $\}$
□ $S = \{$ `BIOS`, `DATA`, `ECON`, `MATH`, `PLSC`, `Other` $\}$

Using `major_dist` and some comparison operators, confirm the Complement Rule using a student's major, a.k.a., confirm that the students with `major == "Other"` is equal to the number of students without a named major in the dataset.

Exercise — write your code here:

Hint
Don't forget to look up the Complement Rule if you need.

Hint
Did you type in raw numbers like 500, or did you use `sum()` on your `student_majors` o

Hint
Did you select rows in the data frame using .loc?

Hint
Did you use a comparison operator to find the rows with "Other" majors?

Hint
Did you use a different comparison operator to find the rows with anything but "Other"

Venn Diagrams

Now, consider the remaining three variables, `DATA118`, `DATA119`, and `DATA120`, which are all Booleans describing whether each of the students are enrolled in DATA118, 119, or 120. This data is visualized in the Venn Diagram below.

Now, use this cell, as well as the `student_majors` data frame and the Venn Diagram, to calculate the probabilities described in the questions below.

Exercise — write your code here:

Question 7. What is the probability that a student is enrolled in DATA118?
Answer:
_____

Question 8. What is the probability that a student is enrolled in DATA119?
Answer:
_____

Question 9. True or False: The event that a student is enrolled in DATA118 and the event that a student is enrolled in DATA119 are mutually exclusive.
☐ True
☐ False

If the event that a student is enrolled in DATA118 and the event that a student is enrolled in DATA119 were independent, what would the probability of a student taking DATA118 and DATA119 be? Use the cell below to calculate your response.

Exercise — write your code here:

Question 10. What is the probability of a student taking DATA118 and DATA119, assuming these two events are independent? Round to three decimal places.
Answer:
_____

Question 11. What is the probability of a student taking DATA118 and DATA119, according to the Venn Diagram? Round to three decimal places.
Answer:
_____

Discuss with a neighbor (or on Ed):
Are the answers to the two previous questions different enough to convince you that taking DATA118 and 119 are dependent? How different would they have to be before you started to believe there is some type of relationship between the two classes?

Question 12. According to the Venn Diagram, what is the probability that a student is taking at least one of DATA118, 119, and 120? Round to three decimal places.
Answer:
_____

Contingency Tables

A conditional probability, or the probability of event A occurring given that event B occurred, is defined as

$$P(A|B) = \frac{P(A \ \& \ B)}{P(B)}$$

Although we don't need a contingency table to calculate these kinds of probabilities, they are a helpful way to visualize different events. Recall that `.pivot_table()` is one way to create a contingency table.

Create a pivot table with majors in the rows of the pivot table and whether or not a student is taking DATA118 in the columns. Be sure to use the `"SID"` column as the `values`. You should be counting the students, not taking the means or using any other aggregating function.

Exercise — write your code here:

As it turns out, there are some ways to improve our pivot table. Check out the documentation for `pivot_table()`. Then, add the argument `margins = True` to your line from the previous exercise. Save this new pivot_table as `majors_x_118`.

Exercise — write your code here:

Discuss with a neighbor (or on Ed):
Between the documentation and the output of your last line of code, what does adding `margins = True` do to your pivot table?

Discuss with a neighbor (or on Ed):
Another very similar function that you could use to create a pivot table is `pd.crosstabs()`. Check out the documentation to see how you might be able

to use it.

Now let's focus on finding the probability that a student is taking DATA118 conditioned on their major. Use this cell, as well as `majors_x_118`, to calculate the probabilities described in the questions below. Remember that we can use counts to calculate conditional probabilities and that the denominators in $P(A \ \& \ B)$ and $P(B)$ will cancel each other out.

Exercise — write your code here:

Question 13. If you were to calculate the probability that a student is taking DATA118 given that they are a DATA major using the definition of conditional probability as defined above, what is event B?
☐ The event that the student is taking DATA118.
☐ The event that the student is not taking DATA118.
☐ The event that a student is a BIOS major.
☐ The event that a student is a DATA major.
☐ The event that a student is included in the data frame.

Question 14. What is the probability that a student is taking DATA118 given that they are a DATA major? Round to three decimal places.
Answer:
_____

Question 15. If you were to calculate the probability that a student is taking DATA118 given that they are a BIOS major using the definition of conditional probability as defined above, what is event A?
☐ The event that the student is taking DATA118.
☐ The event that the student is not taking DATA118.
☐ The event that a student is a BIOS major.
☐ The event that a student is a DATA major.
☐ The event that a student is included in the data frame.

Question 16. What is the probability that a student is taking DATA118 given that they are a BIOS major? Round to three decimal places.
Answer:
_____

Discuss with a neighbor (or on Ed):
Are $P($ `DATA118` $|$ `BIOS`$)$ and $P($ `DATA118` $|$ `DATA`$)$ the same or different? What do you think that says about the independence of the variables `major` and `DATA118`?

Use this cell, as well as `majors_x_118`, to calculate the probabilities described in the questions below.

Exercise — write your code here:

Question 17. If you were to calculate the probability that a student is a DATA major given that they are taking DATA118 using the definition of conditional probability as defined above, what is event B?
☐ The event that the student is taking DATA118.
☐ The event that the student is not taking DATA118.
☐ The event that a student is a BIOS major.
☐ The event that a student is a DATA major.
☐ The event that a student is included in the data frame.

Question 18. What is the probability that a student is a DATA major given that they are taking DATA118? Round to three decimal places.
Answer:
_____

Discuss with a neighbor (or on Ed):
Is $P($ `DATA` $|$ `DATA118` $)$ the same as $P($ `DATA118` $|$ `DATA` $)$? Should they be equal to each other?

Finally, use this cell, as well as your pivot table, to calculate some even more complicated probabilities described in the questions below.

Exercise — write your code here:

Question 19. What is the probability that a student is NOT a DATA major? Round to three decimal places.
Answer:
_____

Question 20. What is the probability that a student is NOT taking DATA118 given that they are NOT a DATA major? Round to three decimal places.
Answer:
_____

Question 21. What is the probability that a student is NOT a DATA major given that they are NOT taking DATA118? Round to three decimal places.
Answer:
_____

Introducing Numeric Variables

So far, we have mostly worked with categorical variables (e.g., a student's major, whether or not they are taking a particular class, etc.). However, we are often more interested in calculating probabilities associated with a **random variable**, where random variables are defined as a variable or process with a numerical outcome.

Create a new column in the `student_majors` data frame representing the number of Introductory Data Science courses (118, 119, 120) a student is taking at this university. Call the new column `num_classes`.

Exercise — write your code here:

Hint
```
## Don't forget that to find the number of introductory DATA courses a student at this uni
```

Hint
```
## You may want to look up the `.sum()` method for data frames.
```

Hint
```
## You looked up `.sum()`, but did you specify the axis correctly?
```

Hint
```
## You looked up `.sum()` and specified the axis correctly, but are you summing the corre
```

Question 22. Describe the sample space of the number of introductory DATA courses a student could be taking at this university.

□ $S = \{0, 1, 2, 3\}$
□ $S = \{0, 1, 2, ..., 497, 498, 499\}$
□ $S = \{$ `major`, `DATA118` `DATA119`, `DATA120` $\}$
□ $S = \{$ `BIOS`, `DATA`, `ECON`, `MATH`, `PLSC` $\}$
□ $S = \{$ `BIOS`, `DATA`, `ECON`, `MATH`, `PLSC`, `Other` $\}$

Discuss with a neighbor (or on Ed):
You might have looked at the column to see what values are in the sample space, but can you use `set()` or `.unique()` to justify your response?

Now use `.groupby()` to find the distribution. Be sure to view on the `"SID"` column, and save your new object as `num_classes_dist`.

Exercise — write your code here:

Use this cell, as well as `num_classes_dist`, to calculate the probabilities described in the questions below.

Exercise — write your code here:

Question 23. What is the probability that a student is not taking any introductory DATA classes? Round to three decimal places.
Answer:
_____

Question 24. True or False: The event that a student is taking two introductory DATA courses and the event that a student is taking three introductory DATA courses are mutually exclusive.
☐ True
☐ False

Question 25. What is the probability that a student is taking two or three introductory DATA courses? Round to three decimal places.
Answer:
_____

Question 26. What is the probability that a student is taking at least one introductory DATA course? Round to three decimal places.
Answer:
_____

Discuss with a neighbor (or on Ed):
Think of at least two ways to calculate the answer from the previous question. It should be the same value as the one from the Venn Diagrams section!

Moving ahead, we will learn how to calculate characteristics like the expected value and variance of a numeric random variable.

Discuss with a neighbor (or on Ed):
What is the expected value of introductory DATA courses a student at our fictional university is taking? How did you arrive at this answer?